

NIKITHA THODUGULI

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EDUCATION

Massachusetts Institute of Technology (June 2027) | Cambridge, MA | GPA: 5.0/5.0

Major: Computer Science & Molecular Biology; Minors: Neuroscience, Political Science

LEADERSHIP

President, MIT Biotech Group (Fall 2026 – present)

Oversee six initiatives and 100+ members; coordinate the annual MBG-Merck Biologics & Vaccines Symposium end-to-end, including speakers, poster presenters, logistics, and sponsorships

Director of Community Engagement, MIT Biotech Group (2024–2026)

Co-launched a \$5K scholarship with Harvard Biotech Club, now in its third year; organized equity-focused fireside chats connecting MBG members with MIT faculty and industry leaders

Projects Team Lead, Global Health Alliance (Fall 2024 – present)

Led a team of 5 to raise funds that erased >\$100K in medical debt at \$1 per \$100 forgiven; developed patient education video outlines on maternal and child health for Village Health Works in Burundi, currently being filmed for use in daily community health workshops

Camp Kesem Counselor (2024–present)

Fundraised >\$500 annually to support two free weeks of sleepaway camp for 200+ Boston-area children affected by a parent's cancer diagnosis; completed counselor training and provided mentorship and emotional support through week-long programming as part of MIT's student-run chapter

8.02 (Physics II) TA (Spring 2024)

Supported a flipped-classroom introductory E&M course by attending lectures, fielding student questions, and leading weekly Friday problem-solving sessions

20.A06 (Biological Engineering) TA (Fall 2024)

Improved the Biobots curriculum within MIT's freshman advising seminar, teaching biofabrication, 3D design, circuits, and neuronal cell and tissue culture

HONORS & AWARDS

Merck Prize, MIT Department of Biology (2025)

Awarded for outstanding research and academic performance in biophysical or bioinformatics sciences

Morais & Rosenblum UROP Award, MIT EECS (2025)

Departmental recognition for undergraduate research excellence

Platform Talk, American Society of Human Genetics Annual Meeting (2025)

Selected from top-scoring abstract submissions for a 10-minute oral presentation, <8% accepted

HKN Member, IEEE Eta Kappa Nu (2025)

Invitation-only honor society for EECS students; juniors in the top 25% of their class are eligible

EXPERIENCE

Summer Research Intern – Harvard Medical School Dept. of Biomedical Informatics (Summer 2026)

Supervisor: Prof. Marinka Zitnik

- Designed a modality-sensitivity benchmark for single-cell multimodal LLMs with a biologically inert baseline, gene order/identity scrambles, and graded marker-gene knockouts ($k = 1-25$) with expression-matched controls to isolate true biological signal from spurious noise
- Scaled the benchmark to a 300k-cell, 35-cell-type atlas via a sharded SLURM pipeline which benchmarks 5 model backends across 2 families (Cell2Sentence, CellWhisperer) using a two-tier, ontology-aware scoring scheme; manuscript for ICLR submission in progress

Undergraduate Researcher – MIT CSAIL (Fall 2023 – present)

Supervisor: Prof. Manolis Kellis

Evolutionary Comparative Genomics [*Winner of Merck Prize '25, EECS Research Award '25*]

- Developed ANCoRA, a damage-aware empirical Bayes algorithm for reference-guided diploid genome assembly, and applied it to build personalized genomes for 34 haplotypes across 17 individuals (5 modern humans, 4 archaic hominins, 8 great apes)
- Applied transformer-based deep learning to predict chromatin accessibility across 100 epigenomic contexts, defining 11k+ lineage-specific cis-regulatory elements (linCREs), which are largely undetected by prior divergence-based methods and enriched in neurodevelopmental pathways and disease loci
- Designed the variant-prioritization scheme that nominated a hominin-gained enhancer of GNB5 (linked to neurodevelopmental disorder), later validated by collaborators via luciferase reporter assays and reciprocal allele-swap experiments; manuscript submitted to *Nature Genetics*
- Presented these findings as an oral platform talk at the 2025 American Society of Human Genetics Annual Meeting, selected from the top 8% of submitted abstracts

Single-Cell Transcriptomics

- Applied the lab's Cell-Projected Phenotypes (CPP) framework to compute cell-level Parkinson's disease pathology scores and identify donors whose transcriptional states concord with case/control labels, addressing disease heterogeneity and enabling mechanistically grounded biomarker analysis

Course Research Project – MIT Deep Learning for Biology (Spring 2026)

- Independently designed and developed a multimodal model for generating cell-state-conditioned chromatin accessibility tracks in the context of Alzheimer's Disease with VAE embeddings for cell state and a discrete diffusion module

Computational Immunomics Intern – Infinitopes (Summer 2025) | Oxford, UK

Supervisor: Dr. Georges Bedran

- Built an end-to-end computational pipeline for generating, co-folding, and analyzing TCR-pMHC complex structures using Boltz-2, a state-of-the-art deep learning sequence-to-structure model
- Designed and executed five benchmarking studies to assess Boltz-2 performance in antigen recognition, structural accuracy, and immune specificity, identifying strengths and limitations of deep learning-based structure prediction for ranking tumor antigen immunogenicity

Genetic Medicines Scholar – Eli Lilly (January 2024) | Seaport, MA

- Pitched a novel genetic therapy for ALS – spanning target identification, drug delivery design, preclinical and clinical study planning, and regulatory approval strategy – as one of 12 MIT undergraduates selected for the inaugural cohort

Undergraduate Researcher – MIT Biological Engineering (June 2024 – December 2024)

Supervisor: Prof. Roger Kamm

- Engineered a neuronal tissue mold in CAD to prevent hypoxic necrotization by incorporating intentional void space for vascular ingrowth, addressing a key bottleneck in brain organoid viability
- Conducted on- and off-chip gelation trials to optimize gelatin-dextran hydrogel composition for synthetic vascular scaffolds in brain organoid models

Computational Neuroscience Researcher – UT Southwestern Biomedical Informatics (Summer 2023)

Supervisor: Prof. Wenhao Zhang

- Simulated Bayesian causal inference for multisensory cue combination in Python via importance sampling and Monte Carlo methods, modeling how the brain integrates visual and auditory stimuli into unified spatial percepts
- Implemented Langevin sampling for Bayesian causal inference, incorporating normalization constants and Gaussian noise to improve posterior estimation

Summer Intern – UT Southwestern Urology (Summer 2022)

Supervisor: Prof. Douglas Strand

- Selected for the UT Southwestern STARS Summer Research Program (< 5% accepted)
- Constructed a cellular anatomy for the female urethra and characterized two previously unknown distinct subtypes of fibroblasts in the female urethra with single-cell RNAseq data
- Performed in situ hybridization with RNAscope to validate the spatial separation of the fibroblast subtypes
- Developed proficiency with key single-cell analysis tools (Seurat, Harmony) for quality control and differential gene expression
- Dissected lower urinary tracts from human donors and sutured tissue in organ bath experiments

SKILLS

Computational & Programming: Python, PyTorch, R, Go, bash, C++, cluster computing, ML on AWS, Git/version control

Machine Learning & Genomics: transformer architectures, diffusion models, VAEs, single-cell analysis (Seurat, Harmony), sequence-to-function modeling

Experimental & Design: neuronal cell culture, brain organoids, biofabrication, tissue engineering, CAD (SolidWorks, Fusion 360), immunohistochemistry, western blot, qPCR, CRISPR, protein purification